

$$b) \forall x \forall y ((x > 0) \wedge (y < 0)) \rightarrow (x - y > 0)$$

For all real numbers x and for all real numbers y , if x is non-negative and y is negative, then the difference $(x - y)$ is positive.

$$c) \forall x \forall y \exists z (x = y + z)$$

For all real numbers x and for all real numbers y , there exists a real number z such that x is equal to the sum $(y + z)$.

d) Let $A(x, y)$ be the statement " x has sent an email message to y ", where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

a) $\exists x \exists y A(x, y)$: From my class, there exists a student x and there also exists a student y s.t. x has sent an email message to y .
Simple: Some student in my class has sent an email message to some other student in my class.

b) $\exists x \forall y A(x, y)$: There exists a student x in my class such that for all students y of my class, x has sent an email message to y .
Simple: Some student in my class has sent an email message to all other students of my class.

c) $\forall x \exists y A(x, y)$: From my class, for all students x , there exists a student y such that x has sent an email message to y .
Simple: Each student in my class has sent an email message to some other student in my class.

d) $\exists y \forall x A(x, y)$: From my class, there exists a student y such that for all student x , x has sent an email message to y .
Simple: Some student in my class has ~~sent~~ ^{received} an email message ~~to~~ ^{from} all the students of my class.

e) $\forall y \exists x A(x, y)$: For all students y of my class, there exists a student x such that x has sent an email message to y .
Simple: Every student of my class has received an email message from some other student of my class.